Summary
Forces acting on an object can result in a change of shape or motion. Having looked at the nature of matter, we can now introduce the idea of fields and forces causing changes. This develops the idea that force interactions between objects can take place even if they are not in contact. Learners should be familiar with forces associated with deforming objects, with stretching and compressing (springs).

Underlying knowledge and understanding
Learners should have an understanding of forces acting to deform objects and to restrict motion. They should already be familiar with Hooke's Law and the idea that, when work is done by a force, it results in an energy transfer and leads to energy being stored by an object. Learners are expected to know that there is a force due to gravity and that gravitational field strength differs on other planets and stars.

Common misconceptions
Learners commonly have difficulty understanding that the weight of an object is not the same as its mass from the everyday use of the term 'weighing'.
The concept of force multipliers can also be challenging even though the basic concepts are ones covered at Key Stage 3.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
PM2.3i	recall and apply: force exerted by a spring (N) = spring constant (N/m) × extension (m)	M1a, M2a, M3a, M3b, M3c, M3d
PM2.3ii	apply: energy transferred in stretching (J) = $\frac{1}{2}$ × spring constant (N/m) × (extension (m))^2	M1a, M2a, M3a, M3b, M3c, M3d
PM2.3iii	recall and apply: gravitational force (N) = mass (kg) × gravitational field strength (N/kg)	M1a, M2a, M3a, M3b, M3c, M3d
PM2.3iv	recall and apply: gravitational potential energy (J) = mass (kg) × gravitational field strength (N/kg) × height (m)	M1a, M2a, M3a, M3b, M3c, M3d
PM2.3v ☑	recall and apply: pressure (Pa) = $\dfrac{\text{force normal to a surface (N)}}{\text{area of that surface (m}^2\text{)}}$	M1a, M2a, M3a, M3b, M3c, M3d
PM2.3vi ☑	recall and apply: moment of a force (N m) = force (N) × distance (m) (normal to direction of the force)	M1a, M2a, M3a, M3b, M3c, M3d

Topic content				Opportunities to cover:		Practical suggestions
Learning outcomes		To include		Maths	Working scientifically	
P2.3a	explain that to stretch, bend or compress an object, more than one force has to be applied	applications to real life situations			WS1.1b, WS1.1e, WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3f, WS1.3g, WS2a, WS2b, WS2c	Use of a liquorice bungee or spring to explore extension and stretching. (PAG P2)
P2.3b	describe the difference between elastic and plastic deformation (distortions) caused by stretching forces				WS1.1b, WS1.1e, WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3f, WS1.3g, WS2a, WS2b, WS2c	Comparisons of behaviour of springs and elastic bands when loading and unloading with weights. (PAG P2)

P2.3c	describe the relationship between force and extension for a spring and other simple systems	graphical representation of the extension of a spring	M1a, M2a, M4a, M4b, M4c	WS1.1b, WS1.1e, WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3f, WS1.3g, WS1.4f, WS2a, WS2b, WS2c	Investigation of forces on springs – Hooke's law. (PAG P2)
P2.3d	describe the difference between linear and non-linear relationships between force and extension		M1a, M2a, M4a, M4b, M4c	WS1.1b, WS1.1e, WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3f, WS1.3g, WS2a, WS2b, WS2c	Investigation of the elastic limit of springs and other materials. (PAG P2)
P2.3e	calculate a spring constant in linear cases		M1a, M2a, M3a, M3b, M3c, M3d		

P2.3f	calculate the work done in stretching		M1a, M2a, M3a, M3b, M3c, M3d, M4a, M4b, M4c, M4f	WS1.1b, WS1.2a, WS1.2b, WS1.2c, WS1.2e, WS1.3a, WS1.3c, WS1.3e, WS1.3f, WS1.3g, WS1.4f, WS2c	Use of data from stretching an elastic band with weights to plot a graph to calculate the work done. (PAG P2)
P2.3g	describe that all matter has a gravitational field that causes attraction, and the field strength is much greater for massive objects				
P2.3h	define weight, describe how it is measured and describe the relationship between the weight of an object and the gravitational field strength, *g*	knowledge that the gravitational field strength is known as *g* and has a value of 10 N/kg at the Earth's surface		WS1.1b	Calculations of weight on different planets.
P2.3i	recall the acceleration in free fall				
P2.3j ☑	apply formulae relating force, mass and relevant physical constants, including gravitational field strength, *g*, to explore how changes in these are inter-related		M1c, M3b, M3c		
P2.3k ☑	describe examples in which forces cause rotation	location of pivot points and whether a resultant turning force will be in a clockwise or anticlockwise direction			

P2.3l ☑	define and calculate the moment of a force	application of the principle of moments for objects which are balanced	M1a, M1c, M2a, M3a, M3b, M3c, M3d	WS1.2a, WS1.2b, WS1.3e, WS2a, WS2b, WS2c	Investigation of moments using a meter ruler, pivot and balancing masses. (PAG P2)
P2.3m ☑	explain how levers and gears transmit the rotational effects of forces	an understanding of ratios and how this enables gears and levers to work as force multipliers	M1c		
P2.3n ☑	recall that the pressure in fluids (gases and liquids) causes a net force at right angles to any surface			WS1.1b, WS1.2a, WS1.4a	Demonstration of balloons being pushed onto a single drawing pin versus many drawing pins.
P2.3o ☑	use the relationship between the force, the pressure and the area in contact	an understanding of how simple hydraulic systems work	M1a, M2a, M3a, M3b, M3c, M3d		